

Homework 1: Boxplots

NAME:

September 15, 2026

Please answer the questions to the best of your ability. Please do not use AI but try your hardest on your own. The point of homework is practice, not getting the correct answer. Please come to office hours or set up a time to meet with me if you have questions!

Question 1

Please answer the following questions using the data....

5, 8, 10, 10, 11, 12, 12, 15, 16

1. Please find the five number summary of the data. That is, please find the minimum, maximum, the median, Q_1 and Q_3 (the first and third quartile).

Min: 5

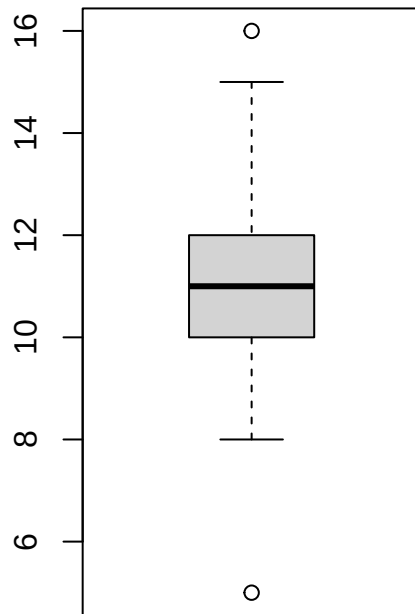
Max: 16

Median: 11

Q_1 = median of the lower half INCLUDING THE MEDIAN when it's in your data so... (5, 8, 10, 10, 11) leads to $Q_1 = 10$

Q_3 = median of the upper half, INCLUDING THE MEDIAN when it's in your data = 12

2. Please draw the associated boxplot with the above data. Please try to draw it to scale



3. Is the boxplot you drew symmetric or skewed? If it's skewed what direction is it skewed?

Roughly symmetric

4. Is there outliers in the data? If so please identify them.

$$\text{IQR} = Q_3 - Q_1 = 2$$

$$1.5 * \text{IQR} = 3$$

$$12 + 3 = 15 \text{ so } 16 \text{ is an outlier}$$

$$10 - 3 = 7 \text{ so } 5 \text{ is also an outlier.}$$

Question 2

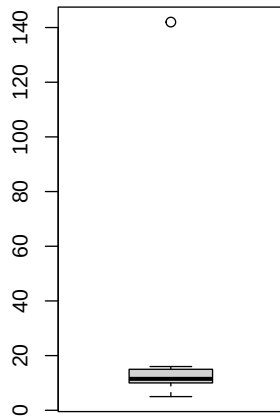
We are now going to change the data that was in question 1 by adding another (very large) data point.

5, 8, 10, 10, 11, 12, 12, 15, 16, 142

1. Please find the five number summary of this new data. That is, please find the minimum, maximum, the median, Q_1 and Q_3 (the first and third quartile).

Min: 5 Max 142 Median: 11.5 Q_1 : 10 Q_3 : 15

2. Please draw the associated boxplot with the new data. Please try to draw it to scale.



3. Is the boxplot you drew symmetric or skewed? If it's skewed what direction is it skewed?

It's very roughly symmetrical but hard to see due to the extreme outlier. Notably the box and the whiskers would imply two different types of skew.

4. Is there outliers in the data? If so please identify them.

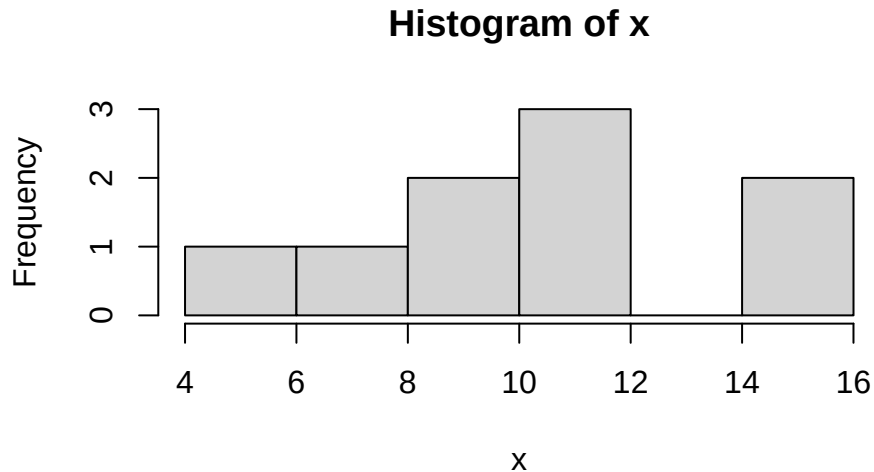
Yes, 142 is an outlier

5. Please comment on how the boxplot you just drew differs from the boxplot in question 1. Please also comment on what is similar.

The lower values (median and below) are roughly the same between the two boxplots. The upper end works differently though. 16 and 5 are no longer outliers but '142 definitely is

Question 3

Please draw a histogram of the data from question 1. I suggest making the x-axis spaced by 2's starting at 0. Please try to make it proportional.



1 Question 4

Shifting topics some now.....Think about our class. If I wanted to split the classroom into subpopulations, what way could I divide the classroom? Basically, suggest ways to split students apart. For example, sex (M vs F) is an easy one. I would like you to propose three different ways (and no, you may not use "sex")

NOTE: Here are some ideas of valid solutions and this is by no means comprehensive

1. Idea 1: Political party (GOP, Democrat, Libertarian, etc...)
2. Idea 2: Athlete Status (Are a student athlete, not a student athlete)
3. Idea 3: In-state/out-of-state students

Populations and Samples

Say the National Park Service is interested in the number of visitors Death Valley gets in a year. Specifically they want to know what proportion of visitors do “backcountry camping”. To look into this, the Park Service randomly sampled 200 visitors to the park and recorded if they did or did not go backcountry camping. Of the 200 visitors talked to, 18 had done backcountry camping while at the park.

1. What is the population the park service is interested in studying?

The population is all visitors to Death Valley NP

2. What parameter (numeric summary) does the park service want to know?

The parameter (a numeric summary of the population) that we are interested in is the proportion of visitors who do backcountry camping.

3. How many people were sampled by the park service?

200 people

4. What statistic did they record from their sample? What was it's numeric value?

The statistic, a numeric summary of the sample, and almost always the same numeric summary as the parameter, is the proportion of visitors in the sample who did backcountry camping. Here it's $18/200$ which is 9%

5. Why do you think the Park Service didn't take a census?

The amount of visitors to the park is too high such that it would have been very costly and energy intensive